PGM Follow — installation and LIVE IN signal qualification

For the engineer setting up the wiring in the OB van. This wiring lets EKO correctly voice replays built from the mixer's PGM — not from the cameras.

1 · The wiring

One router output is dedicated to EKO and connected to SDI 4 · LIVE IN, in place of the classic PGM. The mixer tells the router which input is on air (tally / source follow); the router switches the corresponding source to that output. On this output, **video and VITC switch together** — the mixer's VITC on live, the LSM's VITC as soon as it is taken on air. **The audio never switches**: it stays the mixer's programme output.

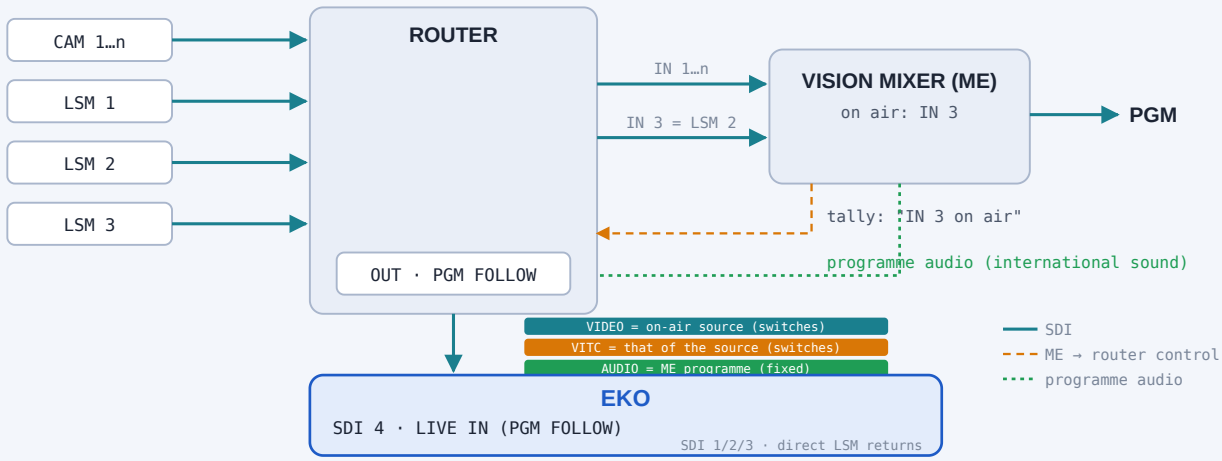
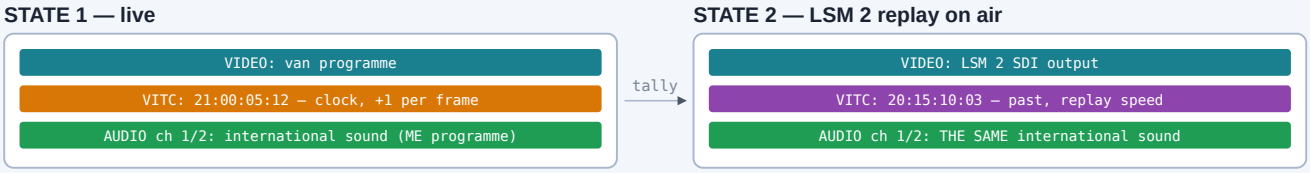


FIG. 1 — ROUTER, MIXER AND PGM FOLLOW OUTPUT TO LIVE IN

2 · The LIVE IN signal, component by component

Component	Live — camera on air	Replay — LSM taken on the ME	Requirement
VIDEO	Van programme output	**The SDI output of the on-air LSM** — the picture being played	REQUIRED · MONITORING QUALITY EKO only computes a signature over a few lines; a low-quality monitoring signal is sufficient.
VITC	Van clock at time of day, undisturbed	**The LSM's VITC**: timecode of the picture played — in the past, at replay speed, including on a freeze	MANDATORY The VITC follows the switched video. During the replay, EKO keeps the continuous clock by extrapolation (+1 frame per frame) from the last live VITC observed.
AUDIO	International sound: ME programme output	**The same international sound, unchanged** — video switching does not affect the audio channels	MANDATORY Must remain the mixer's programme audio output at all times; never replaced by the LSM's audio. Feeds the EKO track without a break.
LSM AUDIO LTC	—	LTC replayed by the LSM on an audio channel, if an LTC was present on its live input	OPTIONAL · NOT REQUIRED In slow motion this LTC is distorted (frequency lowered with speed) and only exists if the LSM outputs audio at reduced speed. EKO does not rely on it.



Classic mistake: letting the audio follow the video → the EKO track breaks at every replay.

FIG. 2 — THE TWO STATES OF LIVE IN

3 · What EKO does with it

When a replay goes on air, EKO sees a VITC jump into the past on LIVE IN. It keeps time-stamping the programme sound with the extrapolated clock and stores, for every aired picture, its real timecode, the programme timecode at which it aired, and its signature. Later, when an LSM replays a clip recorded off the PGM — carrying the recording time, not the picture's — EKO recognises the picture, walks back to the real timecode and generates the right sound. EKO never downgrades a timecode on principle: only recognised pictures are re-timed.

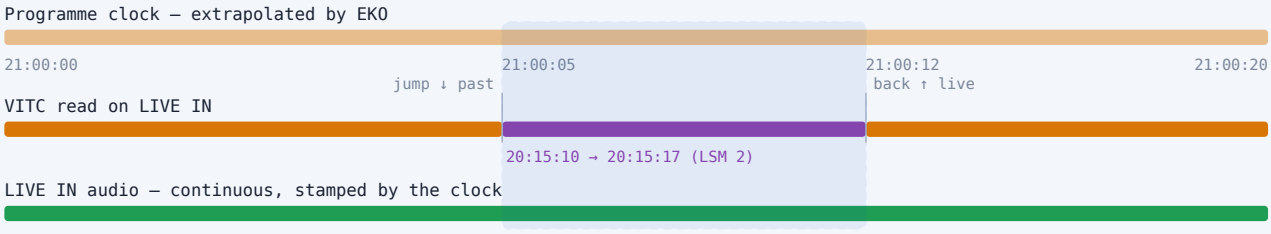


FIG. 3 — VITC JUMP, EXTRAPOLATED CLOCK, MAPPING, RECOGNITION

4 · Installation checks

01 Router: the EKO output follows the on-air source for **video and VITC**; **only the audio** is fixed on the ME programme output.

02 VITC present on the programme output (lines not blanked by the mixer) and on the LSM outputs.

03 LSM taken on the ME: EKO supervision shows the LIVE IN VITC jumping into the past and the LIVE IN sound not moving.

04 Back to live: the VITC rejoins the extrapolated clock with no offset (tolerance: one frame, mixer delay).

05 End-to-end test: record a few seconds of PGM on an LSM while a replay is on air, replay that clip, check that the EKO sound follows the picture and not the recording time.

5 · For reference — classic PGM / PGM Follow

	Classic PGM	PGM Follow
VIDEO	Irrelevant, not read	On-air source; monitoring quality; signature
VITC	Continuous clock, never switches	That of the on-air source: clock on live, LSM VITC on replay
AUDIO	International sound, continuous	**Identical — international sound, continuous, never switches**
AUDIO LTC	Optional, replaces VITC	Optional, not used in slow motion
PGM CLIPS	Voiced at recording time: wrong	Recognised, re-timed to the real timecode: correct